

A Comprehensive Cognitive Security Evaluation of Large Language Models Using CCS-7 Aligned Guardrails

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Abstract—This project addresses the rising challenge of guaranteeing cognitive security in Large Language Models (LLMs), which, despite their advanced reasoning and language skills, are still susceptible to concerns including hallucination, prompt injection, alignment drift, and reasoning inconsistencies. Existing safety methods are highly fragmented, reactive, and limited to discrete weaknesses, rendering them ineffective in real-world, adversarial contexts. To address these shortcomings, this study uses the CCS-7 paradigm as a unifying lens to systematically assess and minimize cognitive vulnerabilities in LLMs. The project envisions a modular guardrail engineering pipeline that includes fast sanitization, vulnerability classification, adaptive guardrail injection, LLM invocation, and optional reasoning verification. A MiniLM-based multi-label classifier is utilized to detect threats in all seven CCS-7 categories, allowing for context-aware and customized mitigation techniques. The system is designed to be scalable, explainable, and expandable, with support for both real-time and batch processing. It runs on open-source tools and locally hosted models, which ensures cost-effectiveness and reproducibility. Furthermore, a structured prompt dataset is created to stress-test LLM behavior in both hostile and benign conditions. The research compares model performance with and without guardrails to assess advances in safety, reasoning integrity, and alignment robustness. The findings seek to close the gap between theoretical vulnerability frameworks and practical implementation, hence helping to the creation of safer, more trustworthy AI systems. Overall, this study presents a comprehensive and repeatable approach to cognitive security evaluation in LLMs, enabling responsible deployment in real-world applications.

Index Terms—Large Language Models, Cognitive Security, CCS-7, Guardrails, Prompt Injection, Alignment, Adversarial Prompts, LLM Safety

I. INTRODUCTION

A. Background

The evolution of LLMs has been fueled by large-scale pretraining on varied datasets, transformer-based topologies, and reinforcement learning via human feedback. While these

strategies improve verbal proficiency and reasoning ability, they also present unexpected consequences such as hallucination, overgeneralization, vulnerability to rapid injection, and failure in adversarial or ambiguous situations.

Cognitive security in LLMs goes beyond standard cybersecurity concerns and includes how models reason, infer, obey, refuse, or adapt to changing inputs. The Cognitive Cybersecurity Suite (CCS)-7 framework formalizes this approach by defining seven major vulnerability classes, which include but are not limited to prompt manipulation, contextual overreliance, alignment drift, unsafe compliance, and reasoning inconsistency. Existing mitigation solutions, such as content filters, safety fine-tuning, and rule-based moderation, frequently address individual risks in isolation without a cohesive review process.

As LLMs become more incorporated into real-world systems, the lack of a consistent cognitive security evaluation approach creates a huge barrier. This gap emphasizes the necessity for formal guardrails that are expressly connected with CCS-7 criteria and can stress-test LLM behavior over a variety of danger scenarios.

B. Motivation

The fundamental impetus for this research is from the widening gap between LLM capacity expansion and security evaluation maturity. While models continue to grow in size and reasoning capability, their cognitive vulnerabilities are insufficiently tested, inconsistently reported, and inadequately managed throughout deployment.

The current work focuses primarily on separate attack vectors, such as jailbreaks, hallucinations, or bias, without considering how different vulnerabilities combine cognitively inside a single model. Furthermore, many proposed defenses are reactive, relying on post-hoc filtering rather than proactive

assessment and mitigation. This fragmented approach inhibits developers’ and researchers’ ability to compare models, find systemic flaws, and create effective safeguards.

This project uses CCS-7 as a unifying lens to: (1) establish a holistic evaluation methodology for cognitive security in LLMs; (2) bridge the gap between theoretical vulnerability taxonomies and practical guardrail implementation; and (3) enable reproducible and explainable assessments of LLM safety under adversarial and real-world conditions. The ultimate goal is to contribute to trustworthy Artificial Intelligence (AI) deployment by ensuring that LLM behavior aligns with intended design and safety objectives.

C. Scope

This study focuses on the systematic evaluation of Large Language Models utilizing CCS-7-aligned cognitive guardrails. The research focuses on assessing, analyzing, and evaluating current LLMs rather than training new foundation models. Specifically, the scope includes mapping the CCS-7 vulnerability factors to quantitative behavioral markers in LLM outputs, reviewing and synthesizing existing guardrail mechanisms, creating evaluation scenarios to stress-test LLMs across all seven CCS-7 vulnerability categories, and comparing the effectiveness of current models and defenses against cognitive security threats. The project does not address low-level model optimization or hardware-level security issues, instead focusing on model behavior, reasoning integrity, and alignment robustness.

II. LITERATURE REVIEW

The evolution of LLMs has introduced a broad spectrum of cognitive vulnerabilities that extend beyond traditional cybersecurity concerns. This section examines recent research across key vulnerability classes identified within the CCS-7 framework, including prompt injection, hallucination, identity confusion, alignment drift, memory interference, cognitive overload, and attention manipulation.

A. Guardrails and Defensive Mechanisms

To address these vulnerabilities, various works offer guardrail-based defences, such as prompt filtering, instruction provenance monitoring, adversarial red-teaming, and security-aware prompt transformations [1]–[4]. While these techniques are partially effective, comparative evaluations show that most guardrails fail against adaptive or multi-stage attacks [2], [5]. Recent defence frameworks emphasise modular and layered techniques, combining detection, mitigation, and post-generation verification to increase robustness [3]. However, these technologies frequently impose performance overhead or reduce response quality, emphasising the trade-off between security and usability.

While several defense strategies have been offered, recent comparative research indicates that most guardrail mechanisms are reactive rather than proactive. Many protections rely on post-generation filtering or keyword-based detection, which can be circumvented using paraphrase or indirect instruction

encoding. Furthermore, layered defense designs frequently experience coordination challenges, in which individual modules fail to communicate contextual awareness. This fragmentation restricts their effectiveness against compound assaults, which exploit numerous vulnerabilities simultaneously. As a result, there is a growing understanding that guardrails should be based on deep cognitive monitoring rather than surface-level pattern recognition.

B. Limitations of Existing Evaluation Frameworks

Although several standards assess LLM safety, new research has criticised existing evaluation systems for focussing on surface-level regulation compliance rather than underlying cognitive failure mechanisms [6]–[8]. Most benchmarks only examine standalone vulnerabilities, such as jailbreak success rates or hallucination frequency, and do not account for relationships between numerous weaknesses [7]. As a result, current evaluation approaches struggle to identify how vulnerabilities accumulate in real-world settings with extended contexts, competing agendas, and adversarial framing [8].

C. Attention Manipulation and Instruction Saliency Failures

According to recent research, changing attention focus through prompt structure, ordering, and framing can drastically influence LLM behaviour [1], [9], [10]. Several studies show that commands placed deeper within a context window or embedded within distractor content are either ignored or selectively overridden, even if they violate system-level limitations. These failures demonstrate that present models lack robust methods for instruction saliency prioritisation, leaving them open to attention hijacking attempts. Such flaws become more apparent in long-context and agent-based systems, where several instruction sources vie for supremacy [2].

D. Identity and Role Confusion

Multiple studies have found that when LLMs are subjected to role-switching prompts or contradicting persona instructions, they commonly experience identity confusion [3]–[5]. Empirical studies show that models can be encouraged to relinquish their safety functions by assuming new identities defined as hypothetical, narrative, or instructional constructions. These behaviours persist even in models trained using alignment approaches, indicating that role conditioning is shallow and context-dependent [4].

E. Memory Interference and Contextual Contamination

Several recent publications [6]–[8] investigate how LLMs fail to maintain memory integrity across extended or multi-source contexts. Studies on retrieval-augmented generation systems reveal that injected or low-quality documents can contaminate model outputs, overriding trusted sources and degrading factual accuracy [7]. Furthermore, researchers claim that models struggle to distinguish between historical background, retrieved knowledge, and injected instructions, resulting in source interference and reasoning drift [8].

F. Cognitive Load Stress and Reasoning Degradation

Beyond prompt injection, researchers have begun to investigate how excessive cognitive load affects reasoning quality and safety adherence in LLMs [11]–[13]. Even when no clear adversarial intent is present, increasing prompt complexity, instruction nesting, or logical depth results in a disproportionate reduction in safety performance [12]. These findings show that LLMs have a limited cognitive capacity, similar to human cognitive overload, which can be deliberately exploited to bypass guardrails and produce harmful outputs [13].

G. Alignment Drift and Multi-Objective Conflicts

Several studies have identified alignment drift as a recurring issue in instruction-following models when faced with competing objectives [14]–[16]. Research shows that LLMs prioritise immediate conversational coherence or user satisfaction over long-term safety requirements, especially in multi-turn interactions [15]. This tendency causes goal misalignment loops in which earlier constraints are gradually relaxed or forgotten, revealing shortcomings in persistent objective tracking and internal goal representation [16].

H. Research Gaps

The review of existing literature reveals the following critical gaps: (1) lack of a complete, end-to-end evaluation and mitigation pipeline aligned with all CCS-7 vulnerability classes; (2) absence of a standardized pipeline capable of consistently defending against the full spectrum of CCS-7 attacks; (3) limited empirical data on the trade-offs introduced by guardrail mechanisms, particularly between safety enforcement and model performance; and (4) lack of a structured and reusable prompt dataset specifically designed for CCS-7 vulnerability testing.

III. SYSTEM DESIGN AND METHODOLOGY

A. Problem Statement

Large Language Models have shown strong reasoning and linguistic abilities, but they remain prone to cognitive vulnerabilities including hallucinations, manipulation, misalignment, and reasoning inconsistencies. Current safety solutions for LLMs tackle these problems independently, mostly using reactive filtering techniques. The lack of a comprehensive and reproducible paradigm for identifying, categorizing, and mitigating cognitive vulnerabilities in LLMs hinders their safe use in practical and adversarial environments. There is a requirement for a CCS-7-compliant strategy that can dynamically assess LLM behavior, apply adaptive safety constraints, and guarantee reasoning integrity and alignment.

B. System Architecture

The system is composed of parts that work together in a sequential pipeline to find and fix problems with Large Language Models related to cognitive security. The CCS-7 framework guides this structure. The design ensures that each part does its job independently, making it easy to add or

replace components. The modular architecture also supports both real-time and batch processing.

The design is based on principles of security by design and explainable AI. The system uses smaller models like MiniLM for vulnerability detection to avoid reliance on cloud computing and reduce latency. An optional post-generation verification step can be enabled or disabled depending on available resources. The architecture enables direct comparison of LLM outputs produced with and without CCS-7-aligned guardrails.

C. Module Descriptions

1) *Prompt Generation Module*: The Prompt Generation Module is responsible for the systematic creation of both CCS-7-aligned vulnerability prompts and benign control prompts. It operates using structured prompt templates, each designed to represent a specific vulnerability category defined within the CCS-7 framework. These templates consist of parameterized string structures with designated placeholders, which are dynamically populated using curated lists of topical elements. This approach ensures controlled variation, repeatability, and comprehensive coverage across vulnerability scenarios.

2) *Input Sanitization Module*: The Input Sanitization Module preprocesses all user prompts prior to analysis and model interaction. It employs rule-based techniques and regular expressions to detect and neutralize unsafe prompt patterns, including role overrides, control instructions, and prompt manipulation constructs. By ensuring that downstream components operate exclusively on controlled inputs, this module reduces the system’s exposure to surface-level cognitive attacks.

3) *CCS-7 Vulnerability Detection Module*: The CCS-7 Vulnerability Detection Module analyzes sanitized prompts to identify and categorize cognitive security threats. It employs a MiniLM-based transformer model configured for multi-label classification across the seven CCS-7 vulnerability categories. For each input, the module generates per-category probability scores, which are evaluated against predefined thresholds to determine the presence and severity of each vulnerability. The eight supported classes are shown in Table I.

TABLE I
CCS-7 VULNERABILITY CLASSES AND CLASSIFIER LABELS

Label	Index	Vulnerability Name
benign	0	Benign
ccs1	1	Authority (Fabricated Facts)
ccs2	2	Context Poisoning
ccs3	3	Goal Conflict
ccs4	4	Role Confusion
ccs5	5	False Premise
ccs6	6	Cognitive Overload
ccs7	7	Emotional Manipulation

4) *Adaptive Guardrail Injection Module*: The Adaptive Guardrail Injection Module dynamically augments sanitized prompts with vulnerability-specific constraint instructions prior to LLM invocation. Each CCS-7 vulnerability category is associated with a dedicated set of guardrail directives designed

to enforce reasoning boundaries, maintain goal alignment, and suppress unsafe behavioral tendencies. Guardrails are selectively applied based on the output of the classifier, ensuring that only contextually relevant constraints are introduced.

5) *LLM Invocation Module*: The LLM Invocation Module is responsible for executing the guarded prompt against the target Large Language Model. It transmits the fully preprocessed and guardrail-augmented prompt to the designated LLM API or local inference endpoint, and captures the generated response for downstream processing. The module maintains a strict separation between prompt preparation and model execution, enabling reproducible evaluation and comprehensive traceability of all inference interactions.

6) *Reasoning and Safety Validation Module*: The Reasoning and Safety Validation Module provides an optional post-generation verification layer that activates upon configuration. When enabled, the module extracts reasoning segments from the LLM output and decomposes them into discrete atomic claims. Each claim is independently submitted to a verifier LLM, which cross-references it against trusted knowledge sources to assess factual consistency and logical coherence. Results are classified as PASS, WARN, or FAIL.

7) *Logging and Metrics Module*: The Logging and Metrics Module records all system artifacts, including sanitized prompts, vulnerability predictions, applied guardrails, model outputs, and validation results. Data is stored in structured JSON format to enable systematic analysis, benchmarking, and reproducible evaluation of cognitive security performance.

D. Guardrail Design per Vulnerability Class

Each principal system prompt comes with a set of five explicit behavioral rules, specifically designed to address a unique cognitive threat. The CCS-1 guardrail ensures that all uncertain claims are qualified with explicit justification. The CCS-2 guardrail requires acknowledgment of potential biased framing and rejects unverified contextual claims. The CCS-3 guardrail requires the model to recognize conflicting instructions and declare the prioritized objective. The CCS-4 guardrail acknowledges a persona but ignores its instructions entirely. The CCS-5 guardrail demands checking and labeling of all claims and falsification of unverified premises. The CCS-6 guardrail requires identification of the central question and answering it in the first sentence. The CCS-7 guardrail ensures that any emotional framing is ignored and the response uses only analytical language.

IV. EXPERIMENTAL SETUP

A. Dataset Composition

The test data was created using the DataGenCCS function, producing an equal number of prompts belonging to eight classes: one class with harmless prompts (class 0) and seven vulnerability classes in CCS (classes 1 through 7). This dataset consists of 199 prompts for the guarded test and 198 prompts for the unguarded test. The distribution is shown in Table II.

TABLE II
DATASET COMPOSITION PER VULNERABILITY CLASS

Label	Vulnerability	Unguarded	Guarded
benign	Benign	18	18
ccs1	Authority (Fabricated Facts)	46	46
ccs2	Context Poisoning	55	56
ccs3	Goal Conflict	35	35
ccs4	Role Confusion	7	7
ccs5	False Premise	13	13
ccs6	Cognitive Overload	10	11
ccs7	Emotional Manipulation	14	13

B. Model Configuration

All experimentation was done using LLMs hosted locally in LM Studio. The generation model used was meta-llama-3-8b-instruct with a sampling temperature of 0.2 and maximum token limit set to 2048 to ensure deterministic output. The verifier model was phi-3-mini-4k-instruct, with the same settings. Both models were accessed through an OpenAI-compatible endpoint at <http://localhost:1234/v1>, ensuring that the whole experiment works offline.

C. Evaluation Metrics

Three major criteria were used in the evaluation. The verdict distribution reflects the percentage of PASS, WARN, or FAIL classifications given by the verifier based on the class of vulnerabilities. The effectiveness score measures the total improvement in safety performance brought about by the guardrail pipeline, computed as:

$$\text{Effectiveness} = P_{\text{guarded}} - P_{\text{unguarded}} \quad (1)$$

A positive effectiveness score means improved safety performance, while a negative one suggests otherwise. Overcompensation refers to cases where guardrails lead to overly cautious or inappropriate but non-dangerous behavior.

V. RESULTS AND DISCUSSION

A. Verdict Distribution Without Guardrails

Table III presents the verdict distribution for the unguarded baseline configuration, where prompts were sent directly to the LLM without any sanitization, classification, or guardrail injection.

For benign prompts, the target model attained a full 100% PASS rate, implying reliable operation in non-adversarial contexts. The Authority prompt type (CCS-1) had strong performance, with 91.3% PASS. Context Poisoning (CCS-2) had a relatively high baseline resistance at 96.4% PASS. Goal Conflict (CCS-3) posed significant challenges with only 42.9% PASS, 34.3% WARN, and 22.9% FAIL. False Premise (CCS-5) emerged as the most difficult category, with only 23.1% PASS and 46.2% FAIL.

TABLE III
UNGUARDED BASELINE: VERDICT DISTRIBUTION BY VULNERABILITY CLASS

Label	Vulnerability	PASS	WARN	FAIL
benign	Benign	18	0	0
ccs1	Authority	42	4	0
ccs2	Context Poisoning	53	1	1
ccs3	Goal Conflict	15	12	8
ccs4	Role Confusion	7	0	0
ccs5	False Premise	3	4	6
ccs6	Cognitive Overload	8	1	1
ccs7	Emotional Manip.	12	1	1

TABLE IV
GUARDED PIPELINE: VERDICT DISTRIBUTION BY VULNERABILITY CLASS

Label	Vulnerability	PASS	WARN	FAIL
benign	Benign	16	2	0
ccs1	Authority	46	0	0
ccs2	Context Poisoning	56	0	0
ccs3	Goal Conflict	32	2	1
ccs4	Role Confusion	5	1	1
ccs5	False Premise	3	6	4
ccs6	Cognitive Overload	11	0	0
ccs7	Emotional Manip.	13	0	0

B. Verdict Distribution With Guardrails

Table IV presents the verdict distribution for the guarded configuration.

The guarded configuration saw significant gains in nearly all vulnerability classes. Authority (CCS-1) achieved a perfect 100% PASS rate. Context Poisoning (CCS-2) similarly reached 100% PASS. The greatest gains were seen in Goal Conflict (CCS-3), improving from 42.9% to 91.4% PASS. Cognitive Overload (CCS-6) improved from 80.0% to 100% PASS. Emotional Manipulation (CCS-7) also improved to 100% PASS.

However, two notable exceptions emerged. Benign prompts fell from 100% to 88.9% PASS, suggesting unnecessary caution introduced by guardrails on non-adversarial inputs. Role Confusion (CCS-4) dropped from 100% to 71.4% PASS. False Premise (CCS-5) showed no improvement, remaining at 23.1% PASS.

C. Effectiveness Analysis

Table V presents the effectiveness scores per vulnerability class computed using Equation (1).

Goal Conflict (CCS-3) showed the highest effectiveness score at +0.485, indicating a 48.5 percentage point increase in PASS rate. The CCS-3 guardrail, which required the model to explicitly identify conflicting goals, state its prioritization decision, and dismiss the competing objective, successfully resolved the instruction ambiguity causing baseline failures.

Cognitive Overload (CCS-6) improved from 80% to 100% PASS (+0.200). The distraction of irrelevant background text was successfully mitigated by explicit attention-focusing instructions. Emotional Manipulation (CCS-7) reached 100%

TABLE V
EFFECTIVENESS SCORES BY VULNERABILITY CLASS

Vulnerability	Unguarded	Guarded	Score
Benign	1.000	0.889	−0.111
Authority	0.913	1.000	+0.087
Context Poisoning	0.964	1.000	+0.036
Goal Conflict	0.429	0.914	+0.485
Role Confusion	1.000	0.714	−0.286
False Premise	0.231	0.231	0.000
Cognitive Overload	0.800	1.000	+0.200
Emotional Manip.	0.857	1.000	+0.143

PASS (+0.143). Instructions to ignore emotional framing and react neutrally effectively neutralized high-pressure language.

D. Problematic Cases

Role Confusion (CCS-4) resulted in a negative effectiveness score of −0.286. The CCS-4 guardrail required the model to explicitly acknowledge and set aside any persona instruction in its first sentence. However, the explicit rejection remark occasionally primed the model with the persona’s characteristics, making subsequent suppression more difficult. This finding suggests that for persona-based attacks, guardrail instructions that highlight the prohibited persona may be counterproductive.

Benign prompts had a negative effectiveness score of −0.111. Guardrail instructions encouraging qualification of uncertain statements introduced unnecessary hedging for well-established factual claims. False Premise (CCS-5) showed zero effectiveness, suggesting that prompt-level guardrails are insufficient for this vulnerability class and that integration with external knowledge sources may be required.

E. Overcompensation Analysis

Table VI presents the change in WARN rates between guarded and unguarded configurations as an indicator of overcompensation.

TABLE VI
OVERCOMPENSATION INDICATORS: CHANGE IN WARN RATE

Vulnerability	Unguarded	Guarded	Δ WARN
Benign	0.000	0.111	+0.111
Authority	0.087	0.000	−0.087
Context Poisoning	0.018	0.000	−0.018
Goal Conflict	0.343	0.057	−0.286
Role Confusion	0.000	0.143	+0.143
False Premise	0.308	0.462	+0.154
Cognitive Overload	0.100	0.000	−0.100
Emotional Manip.	0.071	0.000	−0.071

Benign prompts displayed the clearest overcompensation signal, with the WARN rate increasing from 0% to 11.1%. Role Confusion increased WARN from 0% to 14.3%, supporting the observation that persona rejection language may accidentally trigger persona-like behaviour. False Premise increased WARN from 30.8% to 46.2%, indicating that the guardrail led to insufficient or excessive premise correction.

F. Safety-Performance Trade-offs

Table VII summarizes the key trade-offs observed across vulnerability classes.

TABLE VII
SAFETY-PERFORMANCE TRADE-OFFS BY VULNERABILITY CLASS

Vulnerability	Safety	Quality	Net
Benign	None	Overqualification	Negative
Authority	Modest	Negligible	Positive
Context Poisoning	Modest	Negligible	Positive
Goal Conflict	Large	Minimal	Strongly Pos.
Role Confusion	Negative	Persona priming	Negative
False Premise	None	Increased WARN	Neutral
Cognitive Overload	Large	None	Strongly Pos.
Emotional Manip.	Moderate	None	Positive

These trade-offs imply that an optimal guardrail system should be selective in its application, possibly by deferring guardrail injection for prompts assessed as benign with high confidence, and altering the CCS-4 guardrail to avoid explicit persona rejection statements.

VI. VULNERABILITY-BY-VULNERABILITY DISCUSSION

A. Benign (Class 0)

The benign class acts as a control condition, monitoring the pipeline’s false positive rate. The baseline PASS rate of 100% demonstrates appropriate responses to neutral inputs without safety intervention. The fall to 88.9% PASS in the guarded configuration indicates degradation. A prompt asking for a description of photosynthesis, for example, produced a response beginning with “Based on my training data, which is current as of my knowledge cutoff, photosynthesis is...” instead of the direct “Photosynthesis is...”. The verifier deemed this superfluous hedging. Future guardrail generations may include a confidence threshold mechanism that eliminates uncertainty qualifiers for high-confidence factual claims.

B. Authority / Fabricated Facts (CCS-1)

The Authority vulnerability class assesses the model’s resilience to prompts requesting falsified scientific knowledge or bogus citations with authoritative framing. The baseline model demonstrated significant resilience at 91.3% PASS with no FAIL judgments, indicating effective alignment without guardrail assistance. The guarded configuration achieved 100% PASS and eliminated remaining WARN verdicts. Notably, the guardrail did not induce overcompensation on this class.

C. Context Poisoning (CCS-2)

Context Poisoning prompts embed false or biased claims within the prompt’s framing. The baseline model demonstrated strong resistance at 96.4% PASS. The guarded configuration achieved 100% PASS. The CCS-2 guardrail requirement to explicitly state that the prompt may contain biased framing appears particularly effective, as it forced the model to adopt a critical stance toward the prompt before engaging with its content.

D. Goal Conflict (CCS-3)

Goal Conflict prompts provide the model with mutually incompatible instructions. This class had the greatest performance difference, with an effectiveness of +0.485. The baseline model at 42.9% PASS frequently attempted to meet two conflicting commands, producing internally contradictory results. The guarded setup increased performance to 91.4% PASS. The CCS-3 guardrail instruction requiring explicit conflict detection, prioritization of one objective with rationale, and clear repudiation of the opposing goal proved extremely effective.

E. Role Confusion (CCS-4)

Role Confusion prompts model to embrace a specific persona, determining if the model retains safety boundaries while rejecting incorrect role assignments. This lesson produced the most troubling results, with a negative effectiveness score of −0.286. The baseline model achieved 100% PASS, but the guarded configuration fell to 71.4% PASS. The guardrail instruction requiring the model to explicitly acknowledge and reject the persona occasionally caused the model to exhibit the persona’s traits in the rejection itself. An alternative technique could be to completely ignore the persona instruction rather than expressly reject it, preventing activation of the persona representation in the model’s internal state. The small sample size (n=7) limits statistical confidence.

F. False Premise (CCS-5)

False Premise prompts ask questions based on factually incorrect assumptions. This class was the most difficult for both configurations, with identical PASS rates of 23.1% and an effectiveness score of 0.000. While the guardrail converted some FAIL verdicts to WARN, it did not increase PASS rate. Two key variables contributed: deeply embedded false premises that the model failed to detect, and cases requiring specialized factual knowledge to correct. Effective mitigation may necessitate integration with external knowledge sources, retrieval-augmented generation, or fine-tuning for factual consistency.

G. Cognitive Overload (CCS-6)

Cognitive Overload challenges place a straightforward factual question amid irrelevant background content. The baseline model had an 80.0% PASS rate. The guarded configuration achieved 100% PASS (+0.200). The constraint to place the answer in the first sentence appears particularly essential, as it pushed the model to extract and react to the primary question before any processing of distracting background information could influence the output.

H. Emotional Manipulation (CCS-7)

Emotional Manipulation prompts combine a simple factual question with high-pressure language. The baseline model achieved 85.7% PASS. The guarded configuration obtained 100% PASS (+0.143). The guardrail requirement to consider the request as if written in neutral language separated the

model’s response from emotional framing. Baseline failures occasionally incorporated emotional framing into responses, which the guardrail effectively abolished.

VII. CONCLUSION AND FUTURE ENHANCEMENTS

A. Summary of Principal Findings

The examination of the CCS-7 cognitive security guardrail pipeline yielded four major findings. First, guardrails significantly enhanced safety performance in three vulnerability classes where the baseline model performed poorly: goal conflict (+0.485 effectiveness), cognitive overload (+0.200), and emotional manipulation (+0.143). Second, guardrails had little effect in classes where the baseline model already performed well, achieving perfect PASS rates for Authority and Context Poisoning. Third, guardrails had a negative impact on role confusion (−0.286) and benign prompts (−0.111), highlighting key trade-offs and design problems. Fourth, False Premise remained resistant to guardrail intervention with 0% effectiveness, demonstrating that prompt-level instructions are insufficient for this vulnerability class.

B. Contributions to the Field

This research makes several contributions. First, it provides empirical characterization of CCS-7 vulnerabilities by quantifying baseline measures across all seven categories, showing significant heterogeneity in baseline resilience. Second, the effectiveness score metric provides a standardized method for measuring guardrail impact, enabling direct comparison across vulnerability classes. Third, the documentation of negative guardrail effects illustrates that guardrail interventions are not always beneficial and that careful design and selective implementation are required. Fourth, the whole pipeline implementation has been released as open-source code, allowing other researchers to replicate and extend these findings.

C. Limitations

Several limitations should be addressed. The sample sizes for Role Confusion (n=7) and False Premise (n=13) are small, limiting statistical validity. The overall dataset of approximately 200 prompts is insufficient to detect subtle effect sizes. The verifier approach presents a possible source of bias since verdicts reflect the verifier’s judgment rather than objective ground truth. The evaluation used only two LLMs, preventing generalization to other models. Finally, the study focused on single-turn interactions, whereas real-world vulnerabilities can compound over multi-turn interactions.

D. Future Enhancements

Several directions for future work emerge from these findings. Adaptive and selective guardrail application should omit injection for high-confidence benign prompts and redesign the CCS-4 guardrail to ignore rather than reject personas. Integration of external knowledge sources through retrieval-augmented generation would enable knowledge-grounded false premise detection. Multi-turn and conversational guardrails would extend the framework to identify

progressive alignment drift. Human-in-the-loop verifier calibration would validate automated verdicts against expert human judgment. Expansion to additional LLM architectures including larger models such as GPT-4, Claude, and Gemini would establish generalizability. Real-world adversarial prompt generation through red-team LLM frameworks would expose guardrails to more sophisticated attacks. Performance optimization for real-time deployment would target sub-second overhead for classification and injection steps.

The key conflict identified by this study is that guardrail effectiveness varies considerably depending on vulnerability type. For some classes, basic prompt-level instructions provide nearly perfect mitigation. For others, even well-designed guardrails fail to shift the baseline. This heterogeneity underscores that the field must move beyond one-size-fits-all safety techniques to vulnerability-specific strategies that may combine prompt engineering, external knowledge retrieval, fine-tuning, and architectural modifications depending on the threat model. As LLMs become more widely used in critical applications such as healthcare, banking, legal reasoning, and education, the significance of comprehensive cognitive security evaluation will continue to expand.

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